



Talent Management & AI-Powered Smart Education Platform

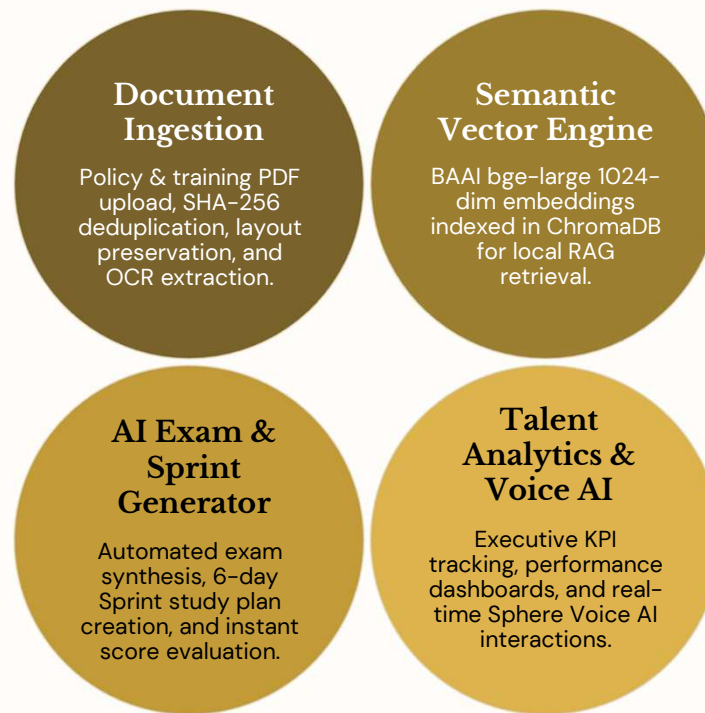
Accelerate workforce capability and career growth through intelligent performance tracking, automated skill assessments, structured 6-day study sprints, and instant semantic search across all internal training documentation.

IQ HOLDERS.

Platform Architecture & Workflow

From Document Ingestion to Career Growth in Four Stages

An end-to-end, privacy-aware AI ecosystem powering enterprise training, semantic document search, automated assessments, and talent analytics.



Every stage operates within a secure environment, transforming raw corporate knowledge into measurable employee competency and career advancement.

TECHNOLOGIES

Engineering the Enterprise Stack

A high-performance, modular stack combining local privacy-aware AI with enterprise talent analytics and interactive voice intelligence.

Flask MVC & Glassmorphism UI

Modern Flask backend with Jinja2 templates, Chart.js analytics, and custom Glassmorphism UI design system.

BAAI/bge-large-en-v1.5

High-density Sentence-Transformers model generating 1024-dimensional embeddings for semantic retrieval.

Groq LLM Engine

LLM engine powering local context-aware RAG, automated AI exam synthesis, and 6-day study plan generation.

SQLite Relational Database

Multi-table relational database storing user credentials, facial descriptors, exam templates, and 6-day study sprints.

ChromaDB Vector Database

Embedded vector database for indexing dense document embeddings and metadata chunks.

Sphere Voice AI & Tool Engine

Real-time Web Speech API assistant with TTS audio readout and dynamic backend tool calling for analytics.

ARCHITECTURE

RETRIEVAL

Chunking, Embeddings & Retrieval Architecture

Chunking Strategy

1000-character chunks with 200-character overlap, plus source pointers and SHA-256.

Embeddings

BAAI bge-large-en-v1.5 creates 1024-d, L2-normalized vectors with versioned metadata.

Vector Store & Retrieval

Local ChromaDB stores vectors and metadata. Top-K cosine search returns ranked chunks with provenance.





Multi-Role Talent Portal & Announcement System

A role-aware portal supports Admin, Manager, and Trainee accounts with SQLite-backed storage.

- Admin: system setup, user management, analytics export
- Manager: assign exams, view KPIs, publish announcements
- Trainee: take exams, view results, receive announcements

▢ Announcements can be broadcast or targeted by role, team, or cohort.



AI Exam Creation & Auto-Grading

The exam engine ingests PDFs, generates assessments, assigns them, and grades automatically where possible.

01

Ingest & Extract:
Parse PDFs and embed content.

03

Assign & Proctor:
Schedule exams with time limits and logging.

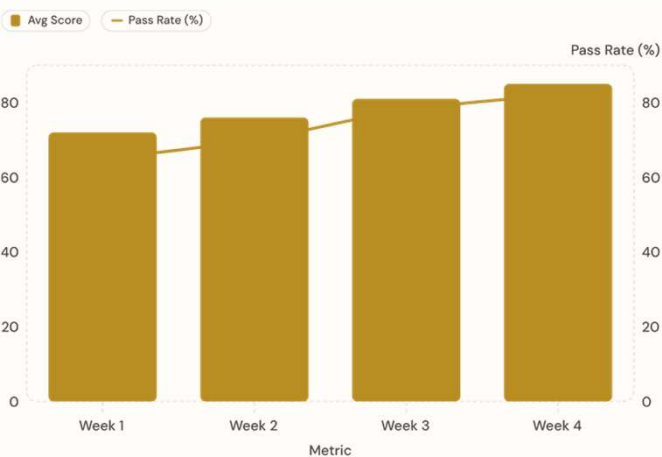
02

Generate Questions:
Create MCQs, short answers, and code prompts.

04

Grade & Review:
Use rule-based scoring and human review for edge cases.

Executive Analytics: KPI Charts & Performance Insights



Dashboard visualizes trainee score trends (bars) with pass-rate overlay (line). Metrics support cohort filtering, drill-down to question-level item analysis, and export for enterprise reporting.



Score Distribution

Histogram and percentile bands to identify at-risk learners and high performers.

Engagement KPIs

Completion rate, time-on-task, and sequence adherence for predictive success modeling.

Operational Metrics

Exam throughput, auto-grade rate, and announcement reach for SLA and capacity planning.

- Next steps: integrate A/B testing for exam variants, add anomaly detection on performance dips, and enable scheduled executive reports.

Voice-Driven Tool Execution — Actions You Can Say



Fetch Trainee Metrics

"Hey Sphere, show Sam's weekly mastery score."

Returns per-lesson metrics, recent activity, and flags for intervention.



Trigger Exam Creation

"Create a Day-5 gateway exam for Module 3."

Generates question set, rubric, and schedules proctoring rules.

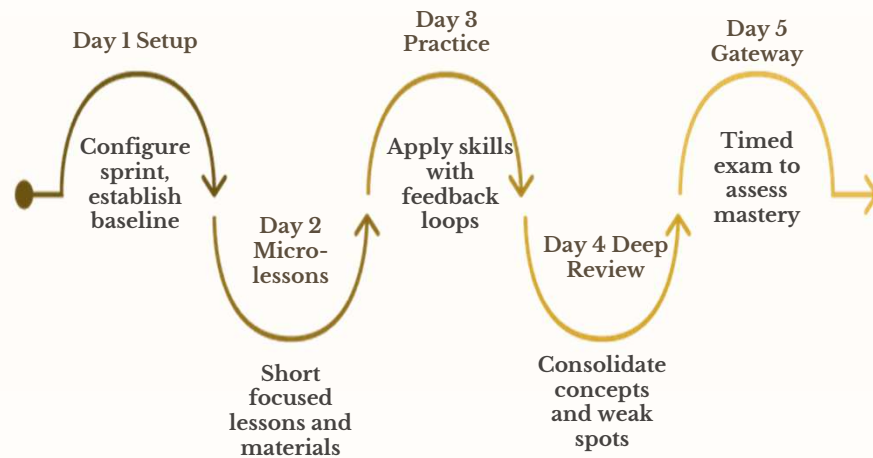


Dispatch Announcements

"Announce sprint kickoff to cohort B." Sends multi-channel notice (in-app, email, voice summary) with delivery trace.

Actions are modeled as idempotent commands with audit logs and RBAC checks. Voice confirmations and undo flows reduce accidental operations.

6-Day AI Study Sprint Orchestrator (with Multilingual Voice Support)



Each sprint is auto-generated by the AI planner: curated daily tasks, prioritized reference materials, suggested time budgets, Day-5 gateway exam creation, and a Day-7 Voice Socratic interview for oral mastery checks.

Multilingual Chat

4-language support with smart selection popups. Voice UI adapts phrasing and TTS voice per locale.

Adaptive Scheduling

Personalized daily task timing based on learner engagement signals and preferred study windows.

Assessment & Gates

Automated Day-5 gating exam plus pass/fail criteria that feed the next sprint's difficulty tuning.

Voice Socratic Interviews

Oral assessments that prompt reasoning, record responses, and provide targeted coach feedback.

- Implementation notes: prioritize low-latency STT for live interviews, use speaker diarization for multi-learner sessions, and retain consented audio for quality review.

System Components & Engineering Highlights



Automated Email Notification Service

Templated SMTP notifications with retry, backoff, and audit logging.

DB-Aware Chatbot Intelligence

Provides controlled read access to trainee records and score tables for live lookups and summaries.

FastAPI + Uvicorn Architecture

Moved to FastAPI on Uvicorn, with CPU-heavy search isolated to keep requests responsive.



SQLite WAL Lock Resilience

WAL mode and connection retry logic reduced Windows file-lock freezes.



Operational Controls

Health checks, circuit breakers, and graceful shutdowns improved runtime stability.

UX, Branding, Release & Next Steps

Finalized UI/UX with glassmorphism cards, accessible light/dark themes, and brand assets for "Talent Management Platform for Employee Performance and Career Growth." Design tokens and component library published.

Release checklist

1

- QA signoff on notifications and chatbot responses
- Load test p95 latency targets met for ASGI service
- DB WAL monitoring in place with alerting
- Brand assets and theme tokens merged into main repo

2

Risk & Mitigation

- Risk: SMTP throttling → Mitigation: queued dispatch with rate limiter
- Risk: DB read hotspots from chatbot → Mitigation: read replicas & caching for summaries
- Risk: Vector search CPU spikes → Mitigation: dedicated workers via ThreadPoolExecutor

Recommended Next Steps

3

1. Run staged rollout to 10% of accounts and monitor delivery & latency
2. Enable read-replica for high-volume analytics queries
3. Hand off branding assets and component docs to product and marketing